Learning a diffusion denoiser by belief propagation: exact scores for Markov data

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Abstract

The global minimiser of the objective that trains a diffusion model is the score of the *empirical* distribution, not of the data distribution — a model attaining it would reproduce its training set and nothing else. Deciding what a network actually learned means comparing it against the score it should have learned, which for realistic data is unavailable. We exhibit a family where it is available and non-trivial: a first-order Markov chain with continuous, non-Gaussian transitions, for which coordinatewise noising leaves the posterior a chain and sum-product returns the exact score. We then drop the assumption that the chain is known and learn it from noised sequences alone: Fisher's identity gives the exact gradient of the marginal likelihood from one sum-product pass, so no differentiation through the recursion is needed. The resulting estimator, with twelve free parameters, attains 8–14$\times$ lower pointwise denoising error than networks trained by denoising score matching on identical data.

1 The problem

Diffusion models are trained by regressing the posterior mean of a clean signal given its noised version [1–4]. A network never sees the data distribution P_0; it sees M samples, and the global minimiser of that regression is the score of the empirical measure $\hat{P}_0 = M^{-1} \sum_\mu \delta(\cdot - \mathbf{a}^\mu)$ smoothed by the forward Gaussian. A model attaining it exactly would generate its training set and nothing else. That diffusion models instead generate new data is the memorisation–generalisation question studied in the statistical mechanics of these models [5, 6].

Progress on it needs a target one can compute. For realistic data the exact score is unavailable, so a network's error cannot be separated from the difficulty of what it was approximating. Tractable families are scarce and mostly Gaussian, where the score is linear and every question about structure has a degenerate answer.

This work. We use a first-order Markov chain: diffusion is applied to sequential data, and studying how structure is exploited requires data that has structure one can dial. The chain is variance-matched, so Gaussian, Laplace, Student, uniform and bimodal innovations share a covariance exactly and differ only beyond second moments. Two things follow — the exact score is computable, and the chain can be *learned* from noised data far more efficiently than a network learns it.

2 The model, and why its score is computable

Forward process. With η standardised Gaussian white noise, $\langle \eta(t)\eta(t')\rangle = \delta(t-t')$, the forward Ornstein–Uhlenbeck process and its solution are

$$\frac{\mathrm{d}\mathbf{x}}{\mathrm{d}t} \;=\; -\mathbf{x} + \sqrt{2}\,\eta(t), \qquad \mathbf{x}(t) \;=\; e^{-t}\mathbf{a} + \sqrt{\Delta_t}\,\mathbf{z}, \qquad \Delta_t = 1 - e^{-2t}, \quad \mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})\,. \quad (1)$$

The reverse-time convention is fixed once in Appendix A and used without variation.

Prior. P_0 factorises as

$$P_0(\mathbf{a}) \;=\; \mu(a_1) \prod_{i=2}^{L} K(a_i \mid a_{i-1}), \qquad K(a' \mid a) \;=\; \varphi(a' - \rho a), \quad (2)$$

with φ an innovation density of mean zero and variance q, and $|\rho| < 1$. We draw $a_1 \sim \mathcal{N}(0,1)$ and set $q = 1 - \rho^2$, which makes $\mathrm{Var}(a_i) = 1$ at every site and $\mathrm{Cov}(a_i, a_j) = \rho^{|i-j|}$ exactly, *whatever the shape of* φ. Variance matching is what makes the family a controlled probe: Gaussian, Laplace, Student, uniform and bimodal innovations share the same covariance and differ only beyond second moments. The chain is covariance-stationary but not strictly stationary; Appendix C states the difference, and shows why it collapses in the Gaussian case.

The score is a posterior mean. From (1), P_t is a Gaussian convolution of P_0. Differentiating under the integral and dividing by $P_t(\mathbf{x})$ — the Gaussian factor over $P_t(\mathbf{x})$ is exactly the posterior density $P(\mathbf{a} \mid \mathbf{x}, t)$ — gives

$$S(\mathbf{x}, t) \;=\; -\frac{\mathbf{x} - e^{-t}\,\langle \mathbf{a} \rangle_{x,t}}{\Delta_t}, \qquad \langle \mathbf{a} \rangle_{x,t} = \int \mathrm{d}\mathbf{a}\;\mathbf{a}\,P(\mathbf{a} \mid \mathbf{x}, t). \quad (3)$$

Equation (3) is a change of variables, not an approximation: computing the score and computing the posterior mean are the same problem. The identity is classical in the empirical-Bayes literature [7–9] and underlies denoising score matching [4, 10]. For squared loss $\langle \mathbf{a} \rangle_{x,t}$ minimises the conditional risk (Appendix B), so it is the reference every estimator below is scored against.

The likelihood factors are *unary*: the forward noise is independent across coordinates, so each observation touches one latent variable. Unary factors reweight node potentials without creating edges among the a_i, so conditioning leaves the factor graph a chain — a tree, on which sum-product returns exact marginals [11–13]. On a chain these are the forward–backward recursions [14, 15], reducing to the Kalman filter when everything is Gaussian [16, 17].

Exactness is at the level of *functional* messages. A message is a function on $\mathbb{R}$, forced by the state space, so an implementation must represent one; we use a truncated grid of N_g points, which makes the recursion exact for a finite-state model whose discretisation error is measured separately. At $N_\mathrm{g} = 401$ it reproduces the Gaussian closed form to 9.2×10^{-15} and brute-force enumeration of the discretised posterior to 1.6×10^{-14}. Cost is $O(LN_\mathrm{g}^2)$ per sequence against $O(N_\mathrm{g}^{L-1})$ for direct summation.

What the standard Gaussian approximation computes. Propagating only the first two moments of each message is the usual approximation. On this model it has an exact characterisation, which turns a heuristic into a measurement:

Proposition 1. *For the chain (2) with linear transitions, Gaussian unary likelihoods and* $\mathbb{E}[\mathbf{a}] = \mathbf{0}$*, moment-projected sum-product returns the LMMSE estimator of the covariance-matched Gaussian model,*

$$\widehat{\mathbf{a}}_{\mathrm{LMMSE}} = \mathrm{Cov}(\mathbf{a}, \mathbf{x})\;\mathrm{Cov}(\mathbf{x})^{-1}\mathbf{x} = e^{-t}\boldsymbol{\Sigma}_0\left(e^{-2t}\boldsymbol{\Sigma}_0 + \Delta_t \mathbf{I}\right)^{-1}\mathbf{x}, \quad (4)$$

for every innovation law with mean zero and variance q*.*

So the gap between moment-projected and full sum-product is precisely the price of replacing the prior by its covariance twin — not an artefact of a message representation. It vanishes as t grows, because noising Gaussianises the posterior, so the shape of φ survives only where the likelihood is weak: on the Laplace chain the median relative score error falls from 0.241 at $t = 0.05$ to 7.3×10^{-6} at $t = 3$ (Figure 1).

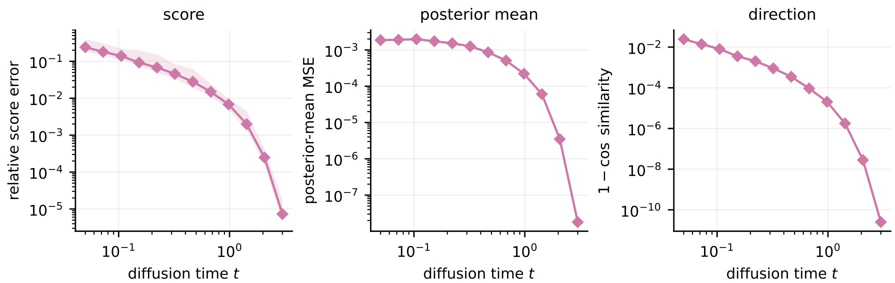

Figure 1: What the Gaussian second-order baseline costs, against sum-product under the true kernel. By Proposition 1 this gap is exactly the price of replacing the prior by its covariance-matched Gaussian, so it measures non-Gaussianity rather than approximation quality. Laplace chain, $\rho = 0.85$; median over seeds with the 10–90 percentile band.

3 Learning the chain from noised sequences

Now treat the kernel as unknown, observe only noised sequences $\mathbf{x}^\mu$, and write $\mathcal{L}(\theta) = \sum_\mu \log P_{t,\theta}(\mathbf{x}^\mu)$. Gradient ascent on $\mathcal{L}$ looks expensive: $\mathcal{L}$ is defined through the recursion, so differentiating it appears to need backpropagation through $2(L-1)$ message updates. It does not.

Proposition 2 (Exact gradient from one sum-product pass)**.** *With* Ξ *the expected transition mass between grid points,*

$$\Xi_{kl} \;=\; \sum_{\mu=1}^{M}\sum_{i=2}^{L} P(a_{i-1}=u_k,\; a_i=u_l \mid \mathbf{x}^\mu)\,, \tag{5}$$

the gradient of the marginal *log-likelihood is*

$$\nabla_\theta \mathcal{L}(\theta) = \sum_\mu \mathbb{E}_\theta[\nabla_\theta \log P_\theta(\mathbf{a}, \mathbf{x}^\mu) \,|\, \mathbf{x}^\mu] = \big\langle \Xi(\theta),\; \nabla_\theta \log K_\theta \big\rangle. \tag{6}$$

Sum-product is not a computation graph to backpropagate through; it supplies the conditional expectation, and one forward–backward pass yields the exact gradient, verified against finite differences of the discretised evidence to $\sim 10^{-9}$. The statistic Ξ is *sufficient* — the complete-data log-likelihood is a sum over edges — so maximisation never revisits the data. For Gaussian innovations that maximisation is closed form, the belief-weighted Yule–Walker equations, and gradient ascent and EM reach the same optimum; gradient ascent needs a tuned step size and EM does not. We estimate the innovation as a mixture, so both the autoregressive coefficient and the innovation density are free: twelve parameters at $C = 4$.

Remark 1 (Information asymmetry)**.** *The estimator is given the Markov factorisation, the kernel's linear-autoregressive form, and homogeneity across sites; the networks are given none of these. What follows measures the value of that structure, not anything about neural denoisers in general.*

Table 1 is the point of the paper in one number: a ratio between 8 and 14 across seven doublings of the training set, against networks trained by denoising score matching in both standard parameterisations, with the parameterisation chosen on a disjoint validation split. The network at the largest budget has not reached the error the estimator attains at the smallest. Markov structure, when the data has it, is worth an order of magnitude in sample efficiency.

4 When structure is invisible to the marginals

A chain makes what is being learned visible in low-order statistics. It need not be. Consider a point starting on a ring of radius ≈1 at a uniformly random angle, taking a rotating random walk; one sample is a whole T-frame trajectory, and the rotation ψ is drawn per trajectory from an unknown $p(\psi)$:

$$z_0 = r_0(\cos\theta_0, \sin\theta_0), \quad r_0 \sim \rho\, e^{-(\rho-1)^2/2\lambda}, \;\; \theta_0 \sim U[0, 2\pi), \qquad z_{u+1} = R_\psi z_u + \sigma\eta_u. \tag{7}$$

Table 1: Relative denoising error against grid BP under the true kernel, averaged over the noise schedule; 16 seeds $\pm$ one standard error, aggregated per seed. Both arms are tuned on a disjoint validation bundle — the network selects its parameterisation *and* training length, EM–BP its iteration count — agreeing with the test-set optimum in 82.2% and 79.5% of cells. EM–BP is more accurate in $1{,}342$ of $1{,}344$. At $M = 2048$ both select their largest allowed budget in 33% and 61% of cells, so that ratio is budget-limited; elsewhere the range is 7.3–12.6.

sequences M	network	EM–BP (12 free)	ratio
32	0.3953 ± 0.0023	$\mathbf{0.0546 \pm 0.0027}$	7.3
64	0.3285 ± 0.0026	$\mathbf{0.0481 \pm 0.0041}$	7.9
128	0.2744 ± 0.0012	$\mathbf{0.0328 \pm 0.0018}$	8.8
256	0.2268 ± 0.0010	$\mathbf{0.0243 \pm 0.0014}$	10.1
512	0.1878 ± 0.0006	$\mathbf{0.0214 \pm 0.0011}$	9.3
1024	0.1569 ± 0.0006	$\mathbf{0.0162 \pm 0.0011}$	12.6
2048	0.1326 ± 0.0005	$\mathbf{0.0134 \pm 0.0008}$	15.7

The rotation is a *gauge*: $U_\psi = \mathrm{blockdiag}(R_{-u\psi})$ is orthogonal, commutes with the isotropic channel, and maps the model onto $\psi = 0$, so $P_t(\mathbf{x} \mid \psi) = P_t^0(U_\psi \mathbf{x})$ is closed form at any T.

Theorem 1 (Marginal blindness)**.** *For the model* (7) *with uniform initial phase, every single-frame marginal is independent of ψ, at every noise level. Consequently a per-frame model carries exactly zero information about the rotation.*

The proof is two lines: $R^{u-k}\eta_k \overset{d}{=} \eta_k$ by isotropy and $R^u z_0 \overset{d}{=} z_0$ by the uniform phase, so $z_u \overset{d}{=} z_0 + \sigma\sqrt{u}\,\xi$, in which ψ does not appear. The marginal is not degenerate — it is a visible, non-Gaussian ring whose radius grows as $\sqrt{u}$. It is simply blind to the dynamics. Radii are not evidence.

This gives the theorem its own control. EM over $p(\psi)$ on the joint likelihood recovers the rotation to $0.15°$ at low noise, degrading to $7.1°$ at $t = 1.4$ as the channel destroys the information. The same EM scored with per-frame marginals cannot move: its likelihood is constant in ψ, so the responsibilities equal $p(\psi)$ and the update is the identity. Across $1{,}344$ cells the blind arm's estimate moves 0.0000 in total variation from its uniform initialisation — a machine-precision zero, which is what a theorem predicting exactly no information requires.

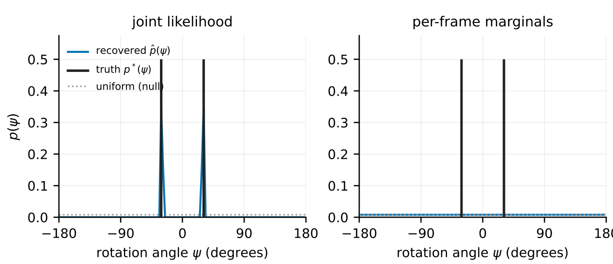

Figure 2: Recovered $p(\psi)$ against the two-species truth $\frac{1}{2}\delta_{+\omega} + \frac{1}{2}\delta_{-\omega}$. Left: EM on the joint likelihood finds both species. Right: the same run scored with per-frame marginals, which by Theorem 1 cannot move off uniform, and does not. Shared y-axis.

5 Scope, and what is open

The estimator assumes exact first-order Markov structure; under a non-Markov prior it is misspecified in a way more data does not repair, and the cost depends on which assumption breaks — a rank-one contamination a chain absorbs, long-range precision coupling it cannot represent. Inference costs $O(LN_{\mathrm{g}}^2)$ per sequence, well above a network forward pass, so the advantage is in data and not wall-clock.

The open question is architectural. Sum-product on a chain suggests a network narrow but as deep as twice the chain, unlike the tree-structured case where the natural depth is the tree's [18, 19]. Which architectures exploit Markovianity, and at what depth, is what this setting is built to measure.

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A Notation

L is the chain length, which is also the ambient dimension; M is the number of observed sequences; N_{g} is the number of quadrature points; K is always the Markov transition kernel, φ the innovation density, ρ the autoregressive coefficient and q its variance. Vectors are bold lowercase, matrices bold uppercase. The forward process is $\mathbf{x}(t) = e^{-t}\mathbf{a} + \sqrt{\Delta_t}\,\mathbf{z}$ with $\Delta_t = 1 - e^{-2t}$; the reverse SDE and its sign conventions follow Anderson [20], Haussmann and Pardoux [21] and are stated in full in the companion paper.

B The score–posterior identity

P_t is a Gaussian convolution of P_0, and $\nabla_{\mathbf{x}} g_{\Delta_t}(\mathbf{x} - e^{-t}\mathbf{a}) = -\frac{\mathbf{x}-e^{-t}\mathbf{a}}{\Delta_t} g_{\Delta_t}(\mathbf{x} - e^{-t}\mathbf{a})$, with the integrand dominated uniformly whenever P_0 has a finite first moment. Dominated convergence then licenses differentiation under the integral; dividing by $P_t(\mathbf{x})$ and recognising the posterior density gives (3). This is Tweedie's formula [7–9]. Since $\langle \mathbf{a} \rangle_{x,t}$ minimises the conditional squared risk, every error below is reported *against* it rather than as raw risk: the Bayes floor is common to all estimators, and subtracting it is what makes the comparison informative.

C Stationarity

With $q = 1 - \rho^2$ and $\mathrm{Var}(a_1) = 1$, second moments propagate exactly: $\mathrm{Var}(a_i) = 1$ and $\mathrm{Cov}(a_i, a_j) = \rho^{|i-j|}$ for every innovation shape. The chain is therefore *covariance*-stationary, which is what makes the variance-matched family a controlled probe. It is not strictly stationary for non-Gaussian φ — the invariant law of $a_i = \rho a_{i-1} + \varepsilon_i$ is that of $\sum_{k\geq 0} \rho^k \varepsilon_{-k}$, which is not Gaussian — and we do not claim it is. The two notions coincide for Gaussians because a Gaussian law is determined by its first two moments.

D Protocol

One frozen configuration, defined in a single file and imported by every experiment so that no run can diverge from it: Laplace chain unless stated, $\rho = 0.85$, $L = 32$, $N_{\mathrm{g}} = 401$, $A = 8$, twelve log-spaced noise levels on $[0.05, 3.0]$, budgets $M \in \{32, \ldots, 4096\}$, and 256 held-out sequences on a disjoint seed. The ring uses $T = 12$. EM stops when the fitted innovation excess kurtosis stops moving, $|\Delta\kappa_4| < 10^{-3}$, never when ρ does: a settled ρ certifies a converged-looking trace over a kernel that is nowhere near converged.